471. Number of members reading Hindi newspaper
$= b + d + f + g = 550 + 100 + 300 + 100 = 1050.$

472. Number of members who read only one newspaper
$= a + b + c = 650 + 550 + 450 = 1650.$

473. Number of members reading no newspaper
$= 2800 - (a + b + c + d + e + f + g)$
$= 2800 - (650 + 550 + 450 + 100 + 200 + 300 + 100)$
$= 2800 - 2350 = 450.$

474. Number of members who read at least 2 newspapers
$= d + e + f + g = 100 + 200 + 300 + 100 = 700.$

475. Required difference $= (e + g) - (d + g) = e - d$
$= 200 - 100 = 100.$

476. Let the consumption of ration be x kg per day. Then,
$75(x + 12) = 90(x + 8)$
$\Leftrightarrow 75x + 900 = 90x + 720$
$\Leftrightarrow 15x = 180 \Leftrightarrow x = 12.$
$\therefore$ Total quantity of ration $= 75(x + 12)$
$= (75 \times 24)$ kg $= 1800$ kg.

Hence, it would last for $\left(\frac{1800}{12}\right)$ days i.e., 150 days.

477. Let l be the length of each candle and x be the number of hours before 4 P.M. when the candles should be lighted.

Length of first candle burnt $= \frac{l}{3} \times x$. Length of second candle burnt $= \frac{l}{4} \times x$.

Remaining length of first candle $= l - \frac{xl}{3}$.

Remaining length of second candle $= l - \frac{xl}{4}$.

$$\therefore\ l - \frac{xl}{4} = 2\left(l - \frac{xl}{3}\right) \Leftrightarrow l\left(1 - \frac{x}{4}\right) = 2l\left(1 - \frac{x}{3}\right)$$

$$\Leftrightarrow 1 - \frac{x}{4} = 2\left(1 - \frac{x}{3}\right) \Leftrightarrow \frac{2x}{3} - \frac{x}{4} = 1$$

$$\Leftrightarrow \frac{5x}{12} = 1 \Leftrightarrow x = \frac{12}{5} = 2\frac{2}{5}.$$

So, the candles should be lighted $2\frac{2}{5}$ hrs i.e., 2 hr 24 min before 4 P.M. i.e., at 1: 36 P.M.

478. Suppose the man bought x apples. Then, number of oranges bought $= (40 - x)$.
Let the cost of each apple be ₹ a and that of each orange be ₹ b.
Then, $ax + b(40 - x) = 17 \Leftrightarrow ax - bx = 17 - 40b$ ...(*i*)
And, $a(40 - x) + bx = 15 \Leftrightarrow ax - bx = 40a - 15$ ...(*ii*)
From (*i*) and (*ii*), we get:
$40a - 15 = 17 - 40b \Leftrightarrow 40a + 40b$
$= 32 \Leftrightarrow 40(a + b) = 32 \Leftrightarrow a + b = \frac{4}{5}.$

Hence, cost of 1 apple and 1 orange

$$= ₹\ \frac{4}{5} = \left(\frac{4}{5} \times 100\right) p = 80\ p.$$

479. Suppose she purchased n books for ₹ 720. Then, cost of each book $= ₹\left(\frac{720}{n}\right)$.

$$\therefore\ \frac{720}{n} - \frac{720}{n+4} = 2 \Rightarrow \frac{(n+4) - n}{n(n+4)}$$

$$= \frac{1}{360} \Rightarrow n^2 + 4n - 1440 = 0$$

$\Rightarrow (n + 40)(n - 36) = 0 \Rightarrow n = 36.$

480. Let the number of pineapple, mango and black-forest pastries be x, y and z respectively.
Then, cost of each pineapple pastry = ₹ x; cost of each mango pastry = ₹ y ;
cost of each black-forest pastry = ₹ z.
Thus, $x + y + z = 23$. And, $x^2 + y^2 + z^2 = 211$.
So, we have to find 3 numbers whose sum is 23 and the sum of whose squares is 211. Such a triplet is (11, 9, 3).

481. Suppose each tube contains x grams initially. Then,

$$4\left[\frac{1}{3}(x+20)\right] = (x-20) + \frac{2}{3}(x+20)$$

$$\Leftrightarrow \frac{2}{3}(x+20) = (x-20)$$

$$\Leftrightarrow \frac{2}{3}x + \frac{40}{3} = x - 20$$

$$\Leftrightarrow \frac{x}{3} = \frac{40}{3} + 20 = \frac{100}{3} \Leftrightarrow x = 100.$$

482. Let the capacity of the first bottle be x litres.
Then, capacity of the second bottle $= 2x$ litres.
And, capacity of the third bottle $= (x + 2x)$ litres
$= 3x$ litres.

Quantity of milk in the first bottle $= \frac{x}{2}$ litres.

Quantity of milk in the second bottle $= \left(\frac{2x}{4}\right)$ litres $= \frac{x}{2}$ litres.

Quantity of milk in the third bottle $= \left(\frac{x}{2} + \frac{x}{2}\right)$ litres $= x$ litres.

$\therefore$ Required fraction $= \left(\frac{x}{3x}\right) = \frac{1}{3}.$

483. Number of passengers after the first stop

$$= \left[540 - \left(\frac{1}{9} \text{ of } 540\right) + 24\right] = 504.$$

Number of passengers after the second stop

$$= \left[504 - \left(\frac{1}{8} \text{ of } 504\right) + 9\right] = 450.$$

484. Let the number of passengers at the start be x.
Then, number of passengers after the first stop

$$= \left(\frac{x}{2} + 125\right).$$

Number of passengers after the second stop

$$= \frac{1}{2}\left(\frac{x}{2} + 125\right) + 100 = \left(\frac{x}{4} + \frac{325}{2}\right).$$

$$\therefore\ \frac{x}{4} + \frac{325}{2} = 250 \Rightarrow \frac{x}{4} = 250 - \frac{325}{2} = \frac{175}{2}$$

$$\Rightarrow x = \left(\frac{175}{2} \times 4\right) = 350.$$